

Test 1 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

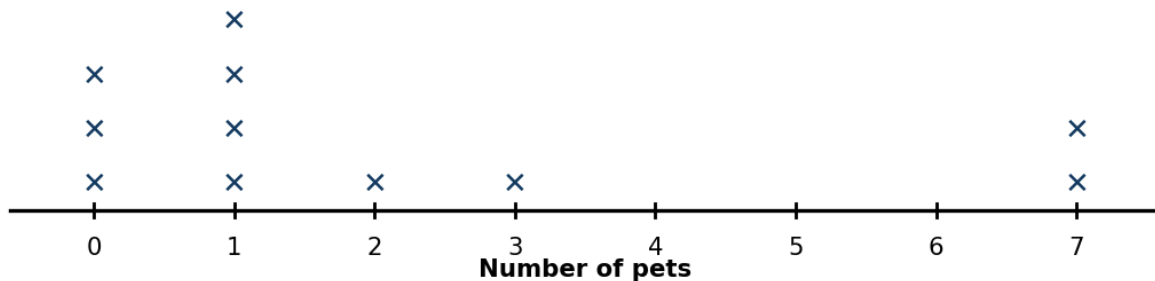
A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. The line plot shows number of pets reported by a group of students.

How many students reported 2 or more?



- A. 7
- B. 6
- C. 5
- D. 4

Show your work

multiple choice

2. Rewrite $4 + 14$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

- A. $2(2 + 8)$
- B. $2(2 + 7)$
- C. $2(9)$
- D. $7(2 + 2)$

Show your work

calculator · multiple choice

3. A data set is shown.

2, 3, 14, 16, 24, 27, 33, 34, 37

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

- A. 2, 24, $17\frac{1}{2}$, $67\frac{1}{2}$, 37
- B. 2, $17\frac{1}{2}$, 16, $67\frac{1}{2}$, 37
- C. 2, 3, 24, 34, 37
- D. 2, $8\frac{1}{2}$, 24, $33\frac{1}{2}$, 37

Show your work

multiple choice

4. Which expression shows the prime factorization of 45?

- A. $3^2 \times 5$
- B. 2×11
- C. $2 \times 3^2 \times 5$
- D. 47

Show your work

calculator · multiple choice

5. Which value of x makes the equation $2x + 3 = 23$ true?

- A. 10
- B. 20
- C. 11
- D. 9

Show your work

multiple choice

6. Andres buys 7 batteries at \$8.51 each and pays a \$2.69 delivery fee.

What is the total cost?

- A. \$27.34
- B. \$78.40
- C. \$59.57
- D. \$62.26

Show your work

calculator · multiple choice

7. Point M is at (1, 1) and point N is at (5, 1).

What is the distance, in units, between point M and point N?

- A. 2
- B. 4
- C. 6
- D. 8

Show your work

multiple choice

8. Which question is a statistical question?

- A. What time does the first bell ring at my school?
- B. How many minutes long is my bus ride to school?
- C. How old is my teacher?
- D. How many text messages did each student send yesterday?

Show your work

multiple choice

9. What value of x makes the equation $x + 2.26 = 5.46$ true?

- A. 3.2
- B. 5.46
- C. 2.26
- D. 7.72

Show your work

multiple choice

10. A data set is shown.

6, 7, 7, 8, 22

The value 26 is added to the data set. What is the mean of the new data set?

- A. 26
- B. $12 \frac{2}{3}$
- C. 10
- D. $15 \frac{1}{5}$

Show your work

multiple choice

11. What is the area of the parallelogram shown, in square feet?



- A. 17
- B. 34
- C. 30
- D. 60

Show your work

multiple choice

12. What is the product of $\frac{3}{5}$ and $\frac{1}{2}$?

- A. $\frac{3}{10}$
- B. $\frac{4}{7}$
- C. $1 \frac{1}{10}$
- D. $1 \frac{1}{5}$

Show your work

calculator · multiple choice

13. A store sells 4 pounds of avocados for \$32.00.

What is the unit price, in dollars per pound?

- A. \$0.50
- B. \$32.00
- C. \$36.00
- D. \$8.00

Show your work

multiple choice

14. Which percent is equivalent to $\frac{1}{4}$?

- A. 400%
- B. 25%
- C. 35%
- D. 2.5%

Show your work

calculator · multiple choice

15. What is the value of $(8)(-4)$?

- A. -32
- B. -28
- C. 4
- D. 32

Show your work

calculator · multiple choice

16. A sixth-grade class at Riviera Middle has 14 students who ride the bus and 18 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

- A. 16:9
- B. 9:7
- C. 7:9
- D. 7:16

Show your work

multiple choice

17. Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -58, -34, -25

Which elevation is farthest from sea level?

- A. -63
- B. -25
- C. -34
- D. -58

Show your work

calculator · multiple choice

18. A parking garage charges more than 20 dollars for the whole day.

Let x represent the cost in dollars. Which inequality represents this situation?

- A. $x < 20$
- B. $x \geq 20$
- C. $x > 20$
- D. $x \leq 20$

Show your work

multiple choice

19. Camila recorded these four numbers in her math journal: -5.5, $-1\frac{3}{4}$, $\frac{3}{4}$, 1.

Which number is the least?

- A. $\frac{3}{4}$
- B. $-1\frac{3}{4}$
- C. 1
- D. -5.5

Show your work

calculator · multiple choice

20. What is the value of $|-15\frac{3}{5}|$?

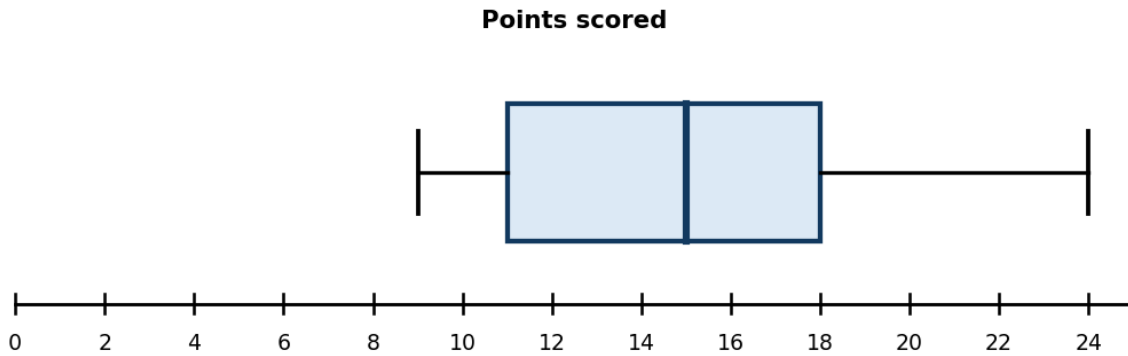
- A. 31.2
- B. -0.06
- C. -15.6
- D. 15.6

Show your work

calculator · multiple choice

21. The box plot shows a set of data.

What is the median of the data?



- A. 15
- B. 18
- C. 11
- D. 17

Show your work

multiple choice

22. What is the solution to the equation $x + 16 = 38$?

- A. 22
- B. 38
- C. 54
- D. 16

Show your work

multiple choice

23. There are 200 students in the sixth grade at Ponce de Leon Middle. 60% of them ride the bus.

How many students ride the bus?

- A. 140
- B. 120
- C. 1200
- D. $3 \frac{1}{3}$

Show your work

multiple choice

24. What is the solution to the equation $9x = 72$?

- A. 72
- B. 63
- C. 8
- D. 648

Show your work

multiple choice

25. What is the value of $13 + (-6)$?

- A. 9
- B. 7
- C. -7
- D. 19

Show your work

calculator · multiple choice

26. A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

- A. 432 inches
- B. 108 inches
- C. 48 inches
- D. 3 inches

Show your work

multiple choice

27. Which expression is equivalent to $2(x + 5)$?

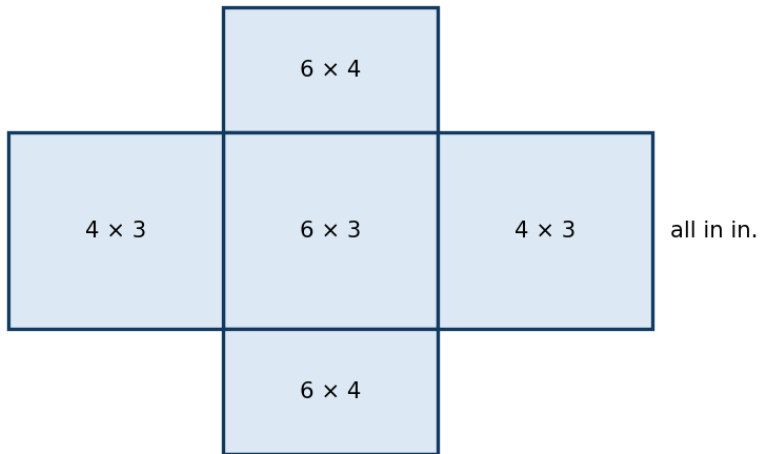
- A. $2x + 5$
- B. $x + 10$
- C. $2x + 10$
- D. $2x + 7$

Show your work

multiple choice

28. The net of a right rectangular prism is shown.

What is the surface area of the prism, in square inches?



- A. 54
- B. 72
- C. 108
- D. 48

Show your work

multiple choice

29. A diver off Virginia Key Beach descends 27 feet below the surface, then rises 17 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

- A. 44
- B. 10
- C. -44
- D. -10

Show your work

calculator · multiple choice

30. What is the value of 10^2 ?

- A. 100
- B. 10
- C. 20
- D. 30

Show your work

calculator · multiple choice

31. The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	6	13	14	17	25	27

- A. 17
- B. 102
- C. 21
- D. 19

Show your work

multiple choice

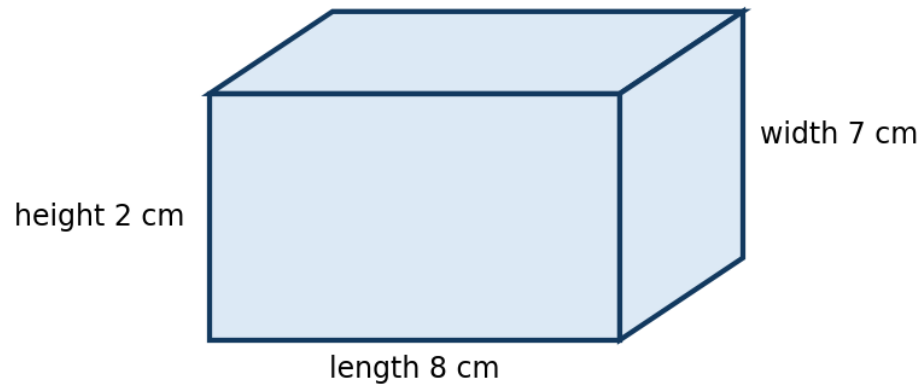
32. What is the greatest common factor of 8 and 20?

- A. 6
- B. 8
- C. 40
- D. 4

Show your work

calculator · multiple choice

33. What is the volume of the right rectangular prism shown, in cubic centimeters?

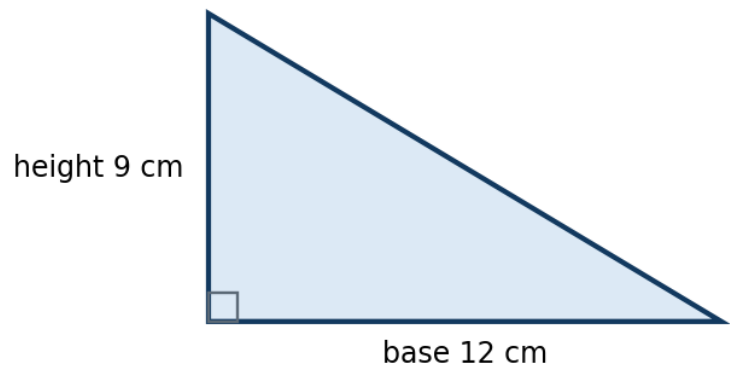


- A. 112
- B. 56
- C. 172
- D. 17

Show your work

multiple choice

34. What is the area of the right triangle shown, in square centimeters?



- A. 108
- B. 10 1/2
- C. 54
- D. 21

Show your work

multiple choice

35. The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
5	6
10	12
10	?

Equivalent ratios

- A. 14
- B. 8
- C. 10
- D. 12

Show your work

multiple choice

36. Use the drop-down menus to place these numbers in order from least to greatest.

-4.5, $-\frac{3}{4}$, $\frac{2}{5}$, 3

Least , , , greatest.

Blank (1) choices: -4.5 | $-\frac{3}{4}$ | $\frac{2}{5}$ | 3

Blank (2) choices: -4.5 | $-\frac{3}{4}$ | $\frac{2}{5}$ | 3

Blank (3) choices: -4.5 | $-\frac{3}{4}$ | $\frac{2}{5}$ | 3

Blank (4) choices: -4.5 | $-\frac{3}{4}$ | $\frac{2}{5}$ | 3

Show your work

calculator · editing task choice

37. What is the value of $5x^2 - 6$ when $x = 8$?

- A. 74
- B. 314
- C. 290
- D. 1594

Show your work

multiple choice

38. What is the product of 21.8 and 0.84?

- A. 1831.2
- B. 18.312
- C. 183.12
- D. 1.8312

Show your work

calculator · multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume